

MYS VASIL'YEVA, Russian Federation

WMO: 322170

Lat: 50.004N

Lon: 155.391E

Elev: 16

StdP: 101.13

Time Zone: 11.00 (E11)

Period: 86-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-10.2	-15.1	1.0	-10.2	-13.7	1.1	-9.2	24.0	-1.7	22.4	-3.3	6.2	280	0.939

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	15.2	12.3	14.0	11.5	12.9	11.4	13.0	14.0	12.3	13.2	11.6	12.5	3.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	9.1	13.3	11.8	8.7	12.4	11.2	8.3	11.8	36.7	14.0	34.9	13.1	33.3	12.6	18.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	18.0	16.0	-14.2	18.7	1.1	1.1	-15.0	19.4	-15.7	20.1	-16.3	20.7	-17.1	21.4
			-14.8	15.4	0.9	1.7	-15.4	16.7	-15.9	17.7	-16.4	18.6	-17.1	19.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-5.4	-4.8	-3.6	-0.9	2.0	5.9	8.9	10.6	9.3	5.8	0.6	-2.8
	DBStd	5.97	2.91	3.08	2.38	1.77	1.60	2.08	1.76	1.35	1.43	1.79	2.60	2.48
	HDD10.0	2903	479	414	423	326	247	126	44	8	29	129	282	396
	HDD18.3	5896	737	648	682	576	505	373	293	238	270	387	532	654
	CDD10.0	48	0	0	0	0	0	2	9	28	9	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.8	8.9	8.0	7.8	7.0	6.2	4.8	4.7	4.9	5.4	7.0	8.4	8.3
Precipitation	PrecAvg	1176	102	91	91	66	61	71	90	104	113	135	141	113
	PrecMax	1364	162	156	164	148	97	137	173	197	179	214	247	205
	PrecMin	915	67	29	47	18	15	12	12	40	71	68	75	48
	PrecStd	113	24	31	28	26	20	35	40	40	30	35	38	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	1.2	1.2	3.1	9.0	17.0	16.2	17.5	15.0	12.0	7.7	3.2
		MCWB	1.0	0.4	0.4	2.2	5.3	11.0	12.9	14.2	11.9	9.2	7.4	2.4
	2%	DB	0.6	0.2	0.2	2.2	6.7	13.7	14.6	15.2	13.8	10.2	6.6	2.1
		MCWB	0.1	-0.1	-0.5	1.4	4.8	9.9	12.3	12.9	11.1	8.3	6.2	1.1
	5%	DB	0.0	-0.3	-0.2	1.7	5.2	11.1	13.0	14.1	12.8	9.4	5.7	1.2
		MCWB	-0.5	-0.6	-0.8	1.0	4.0	8.7	11.4	12.0	11.0	8.0	5.2	0.2
	10%	DB	-0.8	-0.8	-0.5	1.2	4.3	9.1	11.8	13.1	11.8	8.7	4.7	0.7
		MCWB	-1.4	-1.2	-1.1	0.5	3.4	7.5	10.8	11.8	10.5	7.7	4.1	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	0.7	0.8	2.6	6.3	12.0	14.3	14.9	12.8	9.9	7.6	2.9
		MCDB	1.5	0.9	1.0	2.8	7.7	15.5	14.9	16.7	13.7	10.9	7.6	3.2
	2%	WB	0.3	0.0	0.0	1.8	5.1	10.1	12.5	13.5	12.0	9.1	6.4	1.5
		MCDB	0.5	0.1	0.0	1.9	6.4	13.2	13.7	14.6	13.1	9.8	6.5	1.7
	5%	WB	-0.4	-0.5	-0.5	1.2	4.2	8.8	11.6	12.6	11.3	8.5	5.3	0.7
		MCDB	-0.1	-0.3	-0.4	1.4	5.1	10.9	12.8	13.5	12.1	9.0	5.5	1.0
	10%	WB	-1.3	-1.0	-0.9	0.8	3.6	7.7	10.8	12.0	10.7	7.9	4.2	0.0
		MCDB	-0.8	-0.8	-0.7	1.0	4.2	9.0	11.8	12.6	11.4	8.4	4.5	0.5
Mean Daily Temperature Range	5% DB	MDBR	3.6	3.9	4.0	3.0	3.4	4.8	4.5	4.4	4.3	3.9	3.7	3.5
		MCDBR	2.9	3.4	4.4	3.2	5.8	9.6	7.4	6.9	6.4	5.3	4.5	3.6
		MCWBR	3.2	3.2	4.1	2.8	4.2	6.3	5.5	5.0	4.7	4.3	4.4	3.6
	5% WB	MCDBR	2.7	3.3	3.6	2.6	5.2	8.5	6.6	5.5	5.0	4.2	4.1	3.3
		MCWBR	3.1	3.2	3.6	2.5	4.0	5.9	5.2	4.4	4.0	3.9	4.3	3.7
Clear-Sky Solar Irradiance	taub	0.356	0.385	0.437	0.487	0.510	0.483	0.476	0.452	0.393	0.411	0.394	0.362	
	taud	2.092	2.039	1.943	1.868	1.884	2.048	2.128	2.201	2.333	2.102	2.041	2.044	
	Ebn at Noon	629	713	742	752	756	784	783	783	796	675	567	545	
	Edh at Noon	86	116	152	183	188	161	147	130	103	108	90	79	
All-Sky Solar Radiation	RadAvg	0.82	1.43	2.55	3.54	4.16	4.43	4.11	3.58	3.08	1.96	0.99	0.65	
	RadStd	0.04	0.13	0.18	0.33	0.49	0.45	0.46	0.33	0.22	0.15	0.06	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page